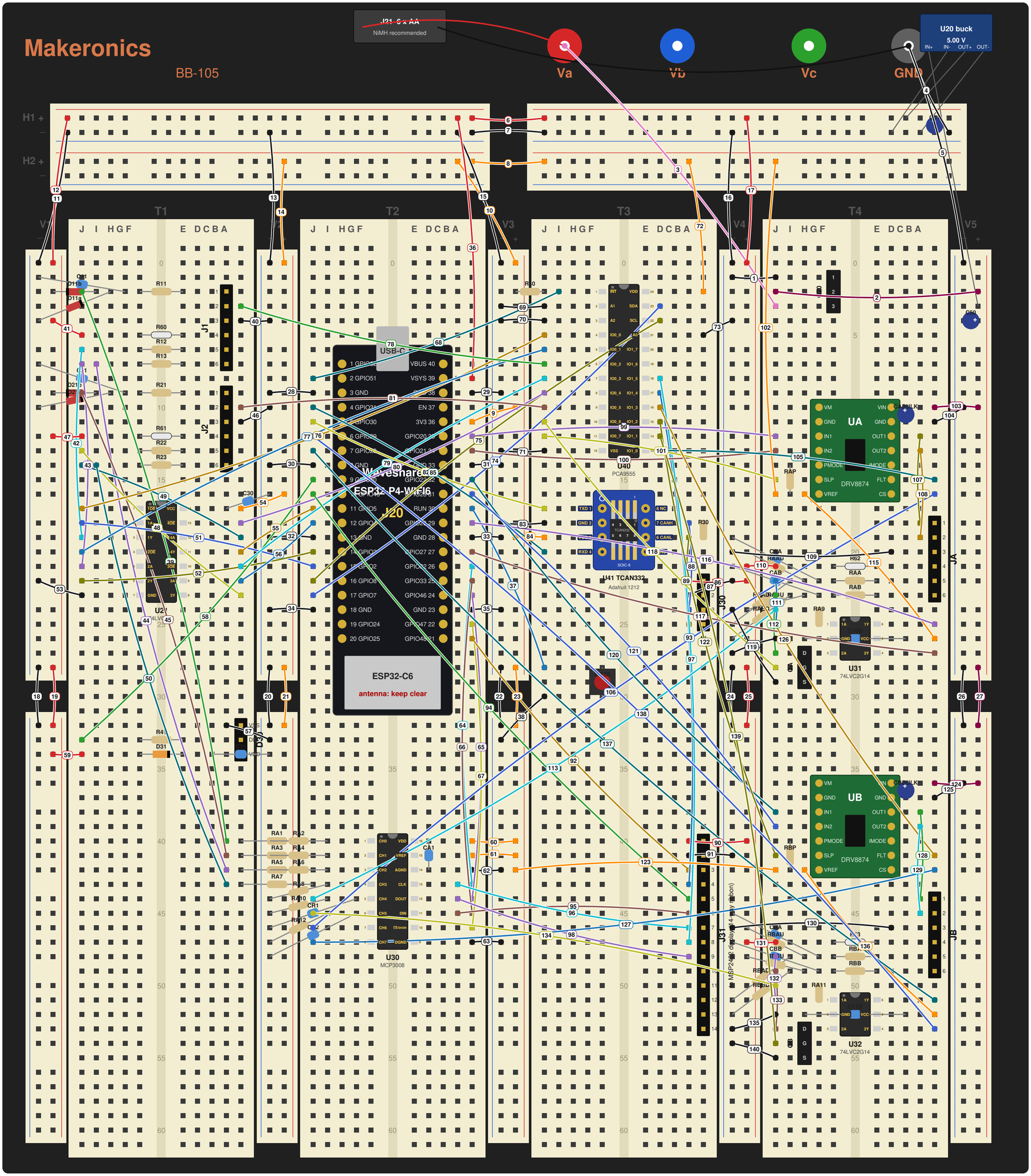
Odin dev carrier on the BB-105 breadboard

Columns read J I H G F | E D C B A from the top end (posts at the top). Rows are the numbers printed on the A-E edge. Numbered tags match the wire list.



Wire colours: GND · +5V · +3V3 · +BATT · VBAT_RAW · other colours: signals

Bus rails are split at the middle; the bridging jumpers are drawn. V5 is motor power only (+BATT / motor GND). Star ground at the GND post.

Bus-strip hole rows are approximate: the bus holes do not line up exactly with the terminal rows. Any free hole on the named rail and half is electrically identical.

Current through breadboard clips: each motor draws up to ~1.2 A (the carrier limit). The VIN and GND wires go straight into the rows the carrier pins sit in, so only one clip carries it, but use short 22 AWG solid wire for the V5 rails, the post wires and the motor OUT1/OUT2 jumpers.

Rails

Rail	Net	Fed from
H1+ (both halves)	+5V	buck OUT+; bridged at the middle
H1− (both halves)	GND (logic)	GND post (star); buck IN−/OUT−
H2+ (both halves)	+3V3	V3+ (from dev board pin 36)
H2−	unused	
V1	− GND, + +5V	H1
V2	− GND, + +3V3	H1−, H2+
V3	− GND, + +3V3	H1−; + from J20 pin 36 (T2-A11)
V4	− GND, + +5V	H1
V5	− motor GND, + +BATT	− from GND post on its own wire; + from Q20 source
post Va	VBAT_RAW	battery + (before reverse protection)
post GND	GND	battery −; the star point

Parts placement

Ref	Value	Holes	Notes
J20	Waveshare ESP32-P4-WIFI6	pins 1-20 at T2-H7..H26, pins 21-40 at T2-C26..C7 (pin n: row 6+n left, row 47-n right)	Plugged straight into T2 as photographed: left pins in H, right in C, pin 1 (GPIO52) and pin 40 (VBUS) on row 7.
J21	6xAA	1=post Va, 2=post GND	Battery pack flying leads: red (+) to post Va, black (-) to post GND.
Q20	DMG2305UX	1=T4-F1, 2=T4-F2, 3=T4-F3	SOT-23 on a 3-pin SIP adapter, pins 1-2-3 top to bottom: G, S, D.
U20	Buck module, 5.00 V	IN+=V5+ r3, IN-=H1− right x60, OUT+=H1+ right x61, OUT-=H1− right x62	Adjustable buck module standing in for U20/L20/R50/R51. Set to 5.00 V BEFORE connecting. Sits off the board edge on flying leads.
C50	10u	1=V5+ r4, 2=V5− r4	+BATT to ground, at the top of V5.
C51	22u	1=H1+ right x63, 2=H1− right x63	Across the +5V rail at the buck output.
SW20	RESET	1=T3-F28, 2=T3-F30	Momentary push button, legs in rows 28 and 30.
J1	Sensor 1	pin 1 at T1-B2 ... pin 6 at T1-B7	Cut EV3 cable on a 6-pin header, pin 1 at the top.
R11	4.7k	1=T1-E2, 2=T1-F2	
C11	330p	1=T1-G2, 2=V1− r1	
D11a	BAT85	K=T1-H2, A=V1− r4	BAT54S low half: cathode to the pin-1 node, anode to GND.
D11b	BAT85	K=V1+ r2, A=T1-I2	BAT54S high half: anode to the pin-1 node, cathode to +5V.
R60	0R	1=T1-F5, 2=T1-E5	0R link (a wire is fine). Load-switch site for hot-plug work.
R12	4.7k	1=T1-E6, 2=T1-F6	
R13	4.7k	1=T1-E7, 2=T1-F7	
J2	Sensor 2	pin 1 at T1-B9 ... pin 6 at T1-B14	Cut EV3 cable on a 6-pin header, pin 1 at the top.
R21	4.7k	1=T1-E9, 2=T1-F9	
C21	330p	1=T1-G9, 2=V1− r7	
D21a	BAT85	K=T1-H9, A=V1− r10	BAT54S low half.
D21b	BAT85	K=V1+ r9, A=T1-I9	BAT54S high half.
R61	0R	1=T1-F12, 2=T1-E12	0R link.
R22	4.7k	1=T1-E13, 2=T1-F13	
R23	4.7k	1=T1-E14, 2=T1-F14	
U21	74LVC125	pin 1 at T1-F17, pins 1-7 down column F, 8-14 up column E	TSSOP-14 on a DIP-14 adapter, notch up.
C30	100n	1=T1-B17, 2=V2− r16	

Ref	Value	Holes	Notes
D30	SK6812 breakout	pin VSS at T1-A32 ... pin VDD at T1-A34	Breakout pins GND / DIN / VDD, top to bottom - match the labels on your module, not this order, if they differ.
R41	470R	1=T1-F33, 2=T1-E33	
D31	1N4148	K=T1-E34, A=T1-F34	Anode to +5V, cathode to the LED supply: drops it to ~4.3 V.
C31	100n	1=T1-C34, 2=V2−r34	
U30	MCP3008	pin 1 at T2-F40, pins 1-8 down column F, 9-16 up column E	DIP-16, notch up, below the dev board.
CA1	100n	1=T2-D40, 2=T2-D42	Vdd row to AGND row.
RA1	10k	1=T1-A40, 2=T2-J40	Spans bus V2, from the T1 landing row to CH0.
RA2	10k	1=T2-I40, 2=V2−r40	
RA3	10k	1=T1-A41, 2=T2-J41	Spans bus V2, from the T1 landing row to CH1.
RA4	10k	1=T2-I41, 2=V2−r41	
RA5	10k	1=T1-A42, 2=T2-J42	Spans bus V2, from the T1 landing row to CH2.
RA6	10k	1=T2-I42, 2=V2−r42	
RA7	10k	1=T1-A43, 2=T2-J43	Spans bus V2, from the T1 landing row to CH3.
RA8	10k	1=T2-I43, 2=V2−r44	
RA10	1M	1=T2-I44, 2=V2−r45	
CR1	10n	1=T2-G44, 2=V2−r46	
RA12	1M	1=T2-I45, 2=V2−r47	
CR2	10n	1=T2-G45, 2=V2−r48	
U40	PCA9555	pin 1 at T3-F2, pins 1-12 down column F, 13-24 up column E	TSSOP-24 on a DIP-24 adapter, notch up. Assumes a 0.3"-wide adapter; see notes.
R40	10k	1=V3+ r2, 2=T3-J2	
U41	TCAN332	pin 1 at T3-F17, pins 1-4 down column F, 5-8 up column E	SOIC-8 on an Adafruit 1212 breakout, "SOIC-8" silkscreen at the bottom. The chip sits turned 90 degrees, dimple top right: legs 1-4 along the top (right to left), 5-8 along the bottom. The board routes them to DIP order: header pin 1 is the hole under the silkscreen circle, top left.
R30	120R	1=T3-A18, 2=T3-A19	Bus termination, CANH row to CANL row.
J30	JST-GH CAN	pin 1 at T3-A22 ... pin 4 at T3-A25	4-pin header for bench use: 5V, GND, CANH, CANL.
J31	MSP2402 display	pin 1 at T3-A40 ... pin 14 at T3-A53	Landing for a 14-way female-to-male jumper ribbon to the display module. Module pin 1 (VCC) on row 40.
UA	Pololu DRV8874 carrier	rows 10-16: logic row in T4-G, power row in T4-D	Straddles the channel, logic row (VM, GND, IN1, IN2, PMODE, SLP, VREF) in G, power row (VIN, GND, OUT1, OUT2, IMODE, FLT, CS) in D, VM/VIN end at the top.
RAP	10k	1=T4-I16, 2=T4-I14	PMODE pull-up to the VREF (+3V3) row.
CABULK	10u	1=T4-C10, 2=T4-C11	VIN row (+) to GND row (-), on the carrier.
JA	Motor A	pin 1 at T4-A18 ... pin 6 at T4-A23	Cut EV3 cable on a 6-pin header, pin 1 at the top.
R62	0R	1=T4-F21, 2=T4-E21	0R link, encoder power from +5V.
RAA	4k7	1=T4-E22, 2=T4-F22	
RAB	4k7	1=T4-E23, 2=T4-F23	
CAA	100p	1=T4-G22, 2=V4−r19	
RAAU	100k	1=V4+ r20, 2=T4-H22	
RAAD	100k	1=V4−r25, 2=T4-I22	
CAB	100p	1=T4-G23, 2=V4−r21	
RABU	100k	1=V4+ r24, 2=T4-H23	
RABD	100k	1=V4−r26, 2=T4-I23	
U31	74LVC2G14	pin 1 at T4-F25, pins 1-3 down column F, 4-6 up column E	SOT-23-6 on a DIP-6 adapter, notch up.
RA9	1M	1=T4-G25, 2=T4-G24	Stand upright: tacho node to the CH4 landing row above it.
C10D	100n	1=T4-C26, 2=T4-H26	VCC row to GND row, arched over the adapter.
QA	2N7000	D=T4-H27, G=T4-H28, S=T4-H29	TO-92, flat face right: D (top), G, S.

Ref	Value	Holes	Notes
UB	Pololu DRV8874 carrier	rows 36-42: logic row in T4-G, power row in T4-D	Straddles the channel, logic row (VM, GND, IN1, IN2, PMODE, SLP, VREF) in G, power row (VIN, GND, OUT1, OUT2, IMODE, FLT, CS) in D, VM/VIN end at the top.
RBP	10k	1=T4-I42, 2=T4-I40	PMODE pull-up to the VREF (+3V3) row.
CBBULK	10u	1=T4-C36, 2=T4-C37	VIN row (+) to GND row (-), on the carrier.
JB	Motor B	pin 1 at T4-A44 ... pin 6 at T4-A49	Cut EV3 cable on a 6-pin header, pin 1 at the top.
R63	0R	1=T4-F47, 2=T4-E47	0R link, encoder power from +5V.
RBA	4k7	1=T4-E48, 2=T4-F48	
RBB	4k7	1=T4-E49, 2=T4-F49	
CBA	100p	1=T4-G48, 2=V4−r45	
RBAU	100k	1=V4+ r46, 2=T4-H48	
RBAD	100k	1=V4−r51, 2=T4-I48	
CBB	100p	1=T4-G49, 2=V4−r47	
RBBU	100k	1=V4+ r48, 2=T4-H49	
RBBD	100k	1=V4−r52, 2=T4-I49	
U32	74LVC2G14	pin 1 at T4-F51, pins 1-3 down column F, 4-6 up column E	SOT-23-6 on a DIP-6 adapter, notch up.
RA11	1M	1=T4-G51, 2=T4-G50	Stand upright: tacho node to the CH5 landing row above it.
C11D	100n	1=T4-C52, 2=T4-H52	VCC row to GND row, arched over the adapter.
QB	2N7000	D=T4-H53, G=T4-H54, S=T4-H55	TO-92, flat face right: D (top), G, S.

Wire list

Numbered as on the layout drawing, in build order. Colour is a suggestion: rails in fixed colours, signals anything.

#	Section	From	To	Net	Colour
1	Power	T4-J1	V4−r1	GND	black
2		T4-J2	V5+ r2	+BATT	dark red
3		post Va	T4-J3	VBAT_RAW	pink
4	Rails	post GND	H1−right x64	GND	black
5		post GND	V5−r0	GND	black
6		H1+ left x31	H1+ right x36	+5V	red
7		H1−left x31	H1−right x36	GND	black
8		H2+ left x31	H2+ right x36	+3V3	orange
9		T2-A11	V3+ r10	+3V3	orange
10		V3+ r0	H2+ left x30	+3V3	orange
11		H1−left x3	V1−r0	GND	black
12		H1+ left x3	V1+ r0	+5V	red
13		H1−left x17	V2−r0	GND	black
14		H2+ left x18	V2+ r0	+3V3	orange
15		H1−left x30	V3−r0	GND	black
16		H1−right x49	V4−r0	GND	black
17		H1+ right x50	V4+ r0	+5V	red
18		V1−r28	V1−r32	GND	black
19		V1+ r28	V1+ r32	+5V	red
20		V2−r28	V2−r32	GND	black
21		V2+ r28	V2+ r32	+3V3	orange
22		V3−r28	V3−r32	GND	black
23		V3+ r28	V3+ r32	+3V3	orange
24		V4−r28	V4−r32	GND	black
25		V4+ r28	V4+ r32	+5V	red
26		V5−r28	V5−r32	GND	black
27		V5+ r28	V5+ r32	+BATT	dark red
28	Dev board	T2-J9	V2−r9	GND	black
29		T2-A9	V3−r9	GND	black
30		T2-J14	V2−r14	GND	black
31		T2-A14	V3−r14	GND	black

#	Section	From	To	Net	Colour
32		T2-J19	V2−r19	GND	black
33		T2-A19	V3−r19	GND	black
34		T2-J24	V2−r24	GND	black
35		T2-A24	V3−r24	GND	black
36		T2-A8	H1+ left x30	+5V	red
37	Reset	T2-A17	T3-J28	ESP_RUN	any signal colour
38		T3-J30	V3−r33	GND	black
39	Sensor port 1	T1-J2	T1-B40	ADC_P1_PIN1	any signal colour
40		T1-A4	V2−r4	GND	black
41		T1-J5	V1+ r4	+5V	red
42		T1-J6	T1-J19	(local)	any signal colour
43		T1-J7	T1-J21	ADC_P1_PIN6	any signal colour
44		T1-I7	T1-B42	ADC_P1_PIN6	any signal colour
45	Sensor port 2	T1-J9	T1-B41	ADC_P2_PIN1	any signal colour
46		T1-A11	V2−r10	GND	black
47		T1-J12	V1+ r12	+5V	red
48		T1-J13	T1-A23	(local)	any signal colour
49		T1-J14	T1-A19	ADC_P2_PIN6	any signal colour
50		T1-I14	T1-B43	ADC_P2_PIN6	any signal colour
51	Sensor buffer U21	T1-J18	T2-J21	UART_P1_TX	any signal colour
52		T1-J22	T2-J20	UART_P1_RX	any signal colour
53		T1-J23	V1−r22	GND	black
54		T1-A17	V2+ r16	+3V3	orange
55		T1-A20	T2-J17	UART_P2_RX	any signal colour
56		T1-A22	T2-J18	UART_P2_TX	any signal colour
57	Status LED	T1-B32	V2−r33	GND	black
58		T1-J33	T2-J15	LED_DATA	any signal colour
59		T1-J34	V1+ r34	+5V	red
60	MCP3008	T2-A40	V3+ r40	+3V3	orange
61		T2-A41	V3+ r41	+3V3	orange
62		T2-A42	V3−r42	GND	black
63		T2-A47	V3−r47	GND	black
64		T2-A21	T2-A43	SPI_SCK	any signal colour
65		T2-A23	T2-A44	SPI_MISO	any signal colour
66		T2-A22	T2-A45	SPI_MOSI	any signal colour
67		T2-A25	T2-A46	SPI_CS_ADC1	any signal colour
68	PCA9555	T2-J8	T3-I2	EXP_INT	any signal colour
69		T3-J3	V3−r3	GND	black
70		T3-J4	V3−r4	GND	black
71		T3-J13	V3−r13	GND	black
72		T3-A2	H2+ right x46	+3V3	orange
73		T3-A5	V4−r4	GND	black
74		T2-J23	T3-D3	I2C_SDA	any signal colour
75		T2-J22	T3-D4	I2C_SCL	any signal colour
76		T1-J17	T3-J5	TXEN_P1	any signal colour
77		T1-J20	T3-J6	RXEN_P1	any signal colour
78		T1-A3	T3-J7	DET_P1	any signal colour
79		T1-A21	T3-J8	TXEN_P2	any signal colour
80		T1-A18	T3-J9	RXEN_P2	any signal colour
81		T1-A10	T3-J10	DET_P2	any signal colour
82	CAN	T2-J11	T3-J17	CAN_TX	any signal colour
83		T3-J18	V3−r18	GND	black
84		T3-J19	V3+ r19	+3V3	orange
85		T2-J10	T3-J20	CAN_RX	any signal colour
86		T3-B22	V4+ r22	+5V	red
87		T3-B23	V4−r22	GND	black
88		T3-B24	T3-B18	CANH	any signal colour

#	Section	From	To	Net	Colour
89		T3-B25	T3-B19	CANL	any signal colour
90	Display	T3-B40	V4+ r40	+5V	red
91		T3-B41	V4−r41	GND	black
92		T2-A26	T3-B42	SPI_CS_DISP	any signal colour
93		T3-D9	T3-B43	DISP_RST	any signal colour
94		T2-J16	T3-B44	DISP_DC	any signal colour
95		T2-B45	T3-B45	SPI_MOSI	any signal colour
96		T2-B43	T3-B46	SPI_SCK	any signal colour
97		T3-D8	T3-B47	DISP_LED	any signal colour
98		T2-B44	T3-B48	SPI_MISO	any signal colour
99	Motor A	T2-A12	T4-J12	MA_IN1	any signal colour
100		T2-A13	T4-J13	MA_IN2	any signal colour
101		T3-J12	T4-J15	MA_SLEEP	any signal colour
102		H2+ right x52	T4-J16	+3V3	orange
103		T4-A10	V5+ r10	+BATT	dark red
104		T4-A11	V5−r10	GND	black
105		T3-D13	T4-A15	MA_FAULT	any signal colour
106		T4-A16	T2-J46	ADC_IA	any signal colour
107		T4-B18	T4-B12	MA_OUT1	any signal colour
108		T4-B19	T4-B13	MA_OUT2	any signal colour
109		T4-D20	V4−r20	GND	black
110		T4-J21	V4+ r21	+5V	red
111	Motor A tacho	T4-J22	T4-J25	ADC_MA	any signal colour
112		T4-J23	T4-J27	(local)	any signal colour
113		T4-J24	T2-J44	(local)	any signal colour
114		T4-J26	V4−r27	GND	black
115		T4-H16	T4-A26	+3V3	orange
116		T4-A25	T2-A18	ENC_A_A	any signal colour
117		T4-A27	T2-A20	ENC_A_B	any signal colour
118		T3-J11	T4-J28	DET_MA	any signal colour
119		T4-J29	V4−r24	GND	black
120	Motor B	T2-A15	T4-J38	MB_IN1	any signal colour
121		T2-A16	T4-J39	MB_IN2	any signal colour
122		T3-D11	T4-J41	MB_SLEEP	any signal colour
123		V3+ r42	T4-J42	+3V3	orange
124		T4-A36	V5+ r36	+BATT	dark red
125		T4-A37	V5−r36	GND	black
126		T3-D10	T4-A41	MB_FAULT	any signal colour
127		T4-A42	T2-J47	ADC_IB	any signal colour
128		T4-B44	T4-B38	MB_OUT1	any signal colour
129		T4-B45	T4-B39	MB_OUT2	any signal colour
130		T4-D46	V4−r46	GND	black
131		T4-J47	V4+ r47	+5V	red
132	Motor B tacho	T4-J48	T4-J51	ADC_MB	any signal colour
133		T4-J49	T4-J53	(local)	any signal colour
134		T4-J50	T2-J45	(local)	any signal colour
135		T4-J52	V4−r53	GND	black
136		T4-H42	T4-A52	+3V3	orange
137		T4-A51	T2-J13	ENC_B_A	any signal colour
138		T4-A53	T2-J12	ENC_B_B	any signal colour
139		T3-D12	T4-J54	DET_MB	any signal colour
140		T4-J55	V4−r54	GND	black

Layout check

gen_breadboard.py resolved the breadboard (row clips, split rails, 140 jumper wires and module-internal ties) into nets and compared them with odin_dev.net: